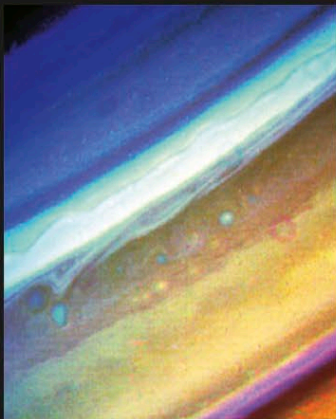


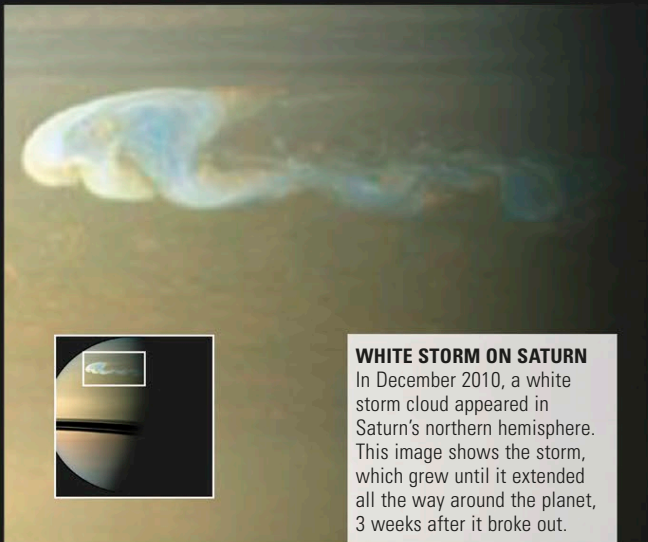


BANDS AND SPOTS

Bands of clouds, spots, and ribbonlike features move across Saturn's visible surface. The spots look small but can be thousands of miles across.

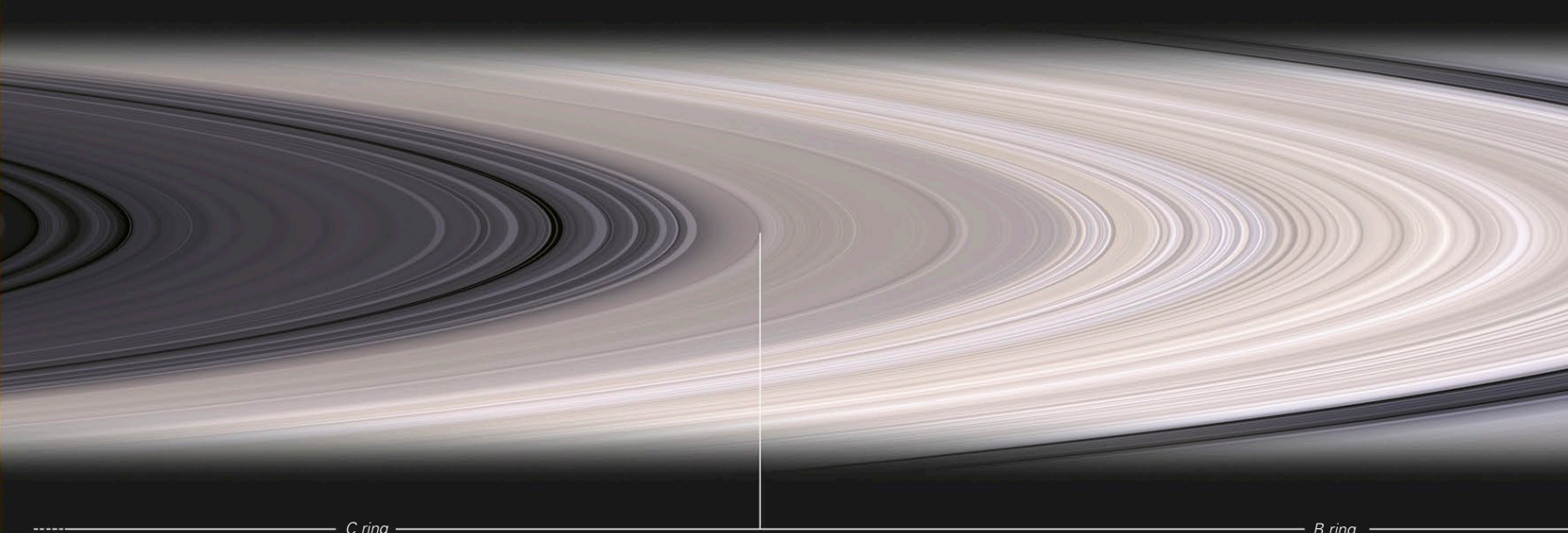
WEATHER

Giant upper-atmosphere storms composed of white ammonia ice can be seen from Earth when they rise through the haze. They occur once every 30 years or so, when it is midsummer in the northern hemisphere, but there is no accepted explanation for the storms yet. The last of these "Great White Spots" was discovered on September 25, 1990. It spread around the planet, almost encircling the equatorial region over about a month. Smaller, different-colored oval spots and ribbonlike features have been observed on a more regular basis. In 2004, Cassini revealed a region then dominated by storm activity, nicknamed "storm alley." Wind speed and direction on the planet are determined by tracking storms and clouds. Saturn's dominant winds blow eastward, in the same direction as the planet's spin. Near the equator, they reach 1,200 mph (1,800 kph).



WHITE STORM ON SATURN

In December 2010, a white storm cloud appeared in Saturn's northern hemisphere. This image shows the storm, which grew until it extended all the way around the planet, 3 weeks after it broke out.



C ring

B ring

MOONS

Saturn has over 80 known moons. Most of these have been discovered since 1980, through exploration by the Voyager and Cassini probes and by improved Earth-based observing techniques. Future observations are expected to confirm the presence of more moons. Titan is the largest and was the first to be discovered, in 1655. It is a unique moon, being the only one in the solar system to have a substantial atmosphere. Saturn's moons are mixes of rock and water ice. Some have ancient, cratered surfaces, and others show signs of resurfacing by tectonics or ice volcanoes. The moons are mostly named after mythological giants. The first to be discovered were named after the Titans, the brothers and sisters of Cronus (Saturn) in Greek mythology. More recent discoveries have Gallic, Inuit, and Norse names.



DIONE

Cassini produced this image of the moon Dione against the backdrop of Saturn's clouds in December 2005. This true-color view reveals the variations in brightness of the moon's icy surface. Dione is Saturn's fourth-largest moon.

SATURN'S MOONS

